Some Harmonic Analysis

— *EYNTKA* Series —

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Contents

Contents **3**

0.1 *Application to Harmonic Analysis 3

0.1.1 Generalization to Locally Compact Abelian Groups 13

0.1.2 The Algebraic-Geometric Meaning of the Dual Group 16

This is a direct extract from my algebra notes that I wanted to make a stand alone blog, see [5].

0.1 *Application to Harmonic Analysis

In this section, we take a brief moment to explore the representation of locally compact abelian groups, such as $\mathbb{R}$. This section will assume some understanding of topological groups, and is mainly to motivate for the reader the connection between the representation theory of this chapter with *Fourier analysis* and more broadly *Harmonic analysis*.

Every group G is always a topological group with the discrete topology (see [5, the appendix on topological groups]). The group G is finite iff it is a compact discrete group. Many groups of interest are either compact (like the torus or compact lie groups), or locally compact (like the real line). The representation theory of these groups rely on the correct framing of the representation in the finite case. For the case of infinite dimension *non-abelian* groups, the build-up is a good deal too complicated for this expository section[1], however many of the major ideas for the abelian are simple enough to exposit, and even better naturally motivate the Fourier transform and abstract harmonic analysis through the lens of representation theory.

Let G be a finite abelian group so that all complex irreducible representations are 1-dimensional, and hence characters and necessarily a homomorphism. As G is finite, say $n = |G|$, then $\chi(g^n) = \chi(e) = 1$ for all $g \in G$, and so $\chi^n = \text{id}$, showing that $\chi : G \to \mathbb{C}^*$ restricts to a map $\chi : S^1$. This shows that all characters are *unitary representations*. All characters for a finite group are necessarily

[1] see [3] for the EYNTKA series on geometric representation theory

unitary representations, while for non-finite groups there can be non-unitary representations[2]. Such groups which have a (general) representation theory that doesn't generalize as well as infinite unitary representations (many representation are indecomposable but not irreducible, for the details see the companion Lie groups notes). Thus, for this sections we stick to maps to S^1 with the natural multiplication structure as the subgroup of $\mathbb{C}^*$: label this collection $\hat{G} = \mathrm{Hom}_{\mathbf{Ab}}(G, S^1)$. As the codomain is a group, $\hom_{\mathbf{Ab}}(G, S^1)$ has a group structure given by point-wise multiplication.

Definition 0.1.1: Pontryagin Group

Let G be locally compact abelian group. Then the *Pontryagin dual* of G is the group:

$$\mathrm{Hom}_{\mathrm{Top}}(G, S^1)$$

with the operation being point-wise multiplication.

This is just another way of representing the characters of the group. Let us stick to the special case where G is a finite abelian group and re-frame the previous theory using the Pontryagin group:

Proposition 0.1.2: Pontryagin Dual

Let G be a finite abelian group, $\hat{G} = \mathrm{Hom}_{\mathbf{Ab}}(G, S^1)$ the group of characters under point-wise multiplication. Then there is an isomorphism:

$$G \cong \widehat{G}$$

Furthermore, there is a natural isomorphism:

$$G \cong \hat{\hat{G}}$$

If G is infinite, it is rarely the case that $G \cong \widehat{G}$ (think $\widehat{L^p} \cong L^q$). Even when G is compact this G need not be isomorphic to $\widehat{G}$. What is most important is that $\widehat{G}$ contains the necessary information to recover G, as the double-dual recovers G. We can think of $\widehat{(-)}$ as an operation that goes back and forth from a group to a unitary representation, and as we are working with unitary representations this collection also forms a group[3]. We can thus imagine going back and forth between these two:

$$\text{Group} \overset{\widehat{(-)}}{\rightleftarrows} \text{Unitary Rep.}$$

***Proof*:**
As G is finite abelian,

$$G \cong \mathbb{Z}/n_1\mathbb{Z} \times \mathbb{Z}/n_2\mathbb{Z}/ \times \cdots \times \mathbb{Z}/n_k\mathbb{Z}$$

where $n_1 \big| n_2 \big| \cdots \big| n_k$. Choose generators, $a_1, ..., a_k$. Define the indicator characters on the

[2] $\mathrm{SL}_2(\mathbb{R})$ acting on $\mathbb{R}^2$ need not be unitary

[3] the collection of representations also form a ring called the *representation ring* and has fascinating ties to K-theory and intersection theory, but it is again too much for this section and the connections are discussed in [1, Part II] of the EYTNKA series

generators:

$$\chi_i(a_j) = \begin{cases} e^{\frac{2\pi i}{n_i}} & i = j \\ 1 & i \neq j \end{cases}$$

These certainly generate $\hat{G}$. Define the map on the generators:

$$\Phi(a_i) = \chi_i$$

This is an isomorphism.

For the double-dual, take the natural map:

$$\Phi(g_i) = \mathrm{ev}_{g_i}$$

where as usual $\mathrm{ev}_{g_i}(\chi) = \chi(g_i)$. This is easy to check that is an isomorphism, completing the proof.

Now, when studying spaces we study all (continuous) functions on a space. In our case, our space is finite abelian groups. Let G have the discrete topology so that the set of continuous functions from G to $\mathbb{C}$ is all set functions. We want a $\mathbb{C}$-algebra isomorphism between $\mathbb{C}[G]$ and $\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C})$. These two sets are clearly bijective via the natural mapping:

$$\sum_i a_i g_i \mapsto (g_i \mapsto a_i)$$

And this map preserves addition if addition is defined point-wise in the hom-set:

$$\begin{aligned}
&\Phi\left(\sum_i a_i g_i + \sum_i b_i g_i\right) \\
&= \Phi\left(\sum_i (a_i + b_i) g_i\right) \\
&\mapsto (g_i \mapsto a_i + b_i) \\
&= (g_i \mapsto a_i) + (g_i \mapsto b_i) \\
&= \Phi\left(\sum_i a_i g_i\right) + \Phi\left(\sum_i b_i g_i\right)
\end{aligned}$$

But it does *not* preserve multiplication if the operation is pointwise multiplication:

$$\begin{aligned}
&\Phi\left(\sum_{g\in G} a_i g \sum_{g\in G} b_i g\right) \\
&= \Phi\left(\sum_{g\in G} \sum_{g_k g_\ell = g} (a_k b_\ell) g\right) \\
&\mapsto (g_i \mapsto \sum_{g_k g_\ell = g_i} (a_k b_\ell)) \\
&\neq (g_i \mapsto a_i b_i) \\
&\neq (g_i \mapsto a_i) \cdot (g_i \mapsto b_i) \\
&= \Phi\left(\sum_i a_i g_i\right) \cdot \Phi\left(\sum_i b_i g_i\right)
\end{aligned}$$

This is because the points g_i act on each other. There is a natural multiplication function defined on $\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C})$ called *convolution*:

Definition 0.1.3: Convolution

Let $p, q \in \mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C})$. Then their *convolution* is define pointwise as:

$$(p * q)(g) = \sum_{g_i g_j = g} p(g_i) q(g_j) = \sum_{h\in G} p(h) q(h^{-1} g)$$

Lemma 0.1.4: Isomorphism With Convolution

Let G be a finite abelian group. Then:

$$(\mathbb{C}[G], +, \cdot) \cong_{k\text{-}\mathbf{Alg}} (\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C}), +, *)$$

where the left hand side is the usual group ring and the right hand side has convolution as multiplication

***Proof*:**
refer to the above calculations.

Now, $(\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C}), +, *)$ is a $|G|$-dimensional $\mathbb{C}$ vector-space with the natural basis given by the indicator function:

$$\delta_h(g) = \begin{cases} 1 & g = h \\ 0 & g \neq h \end{cases}$$

Namely for any $f : G \to \mathbb{C}$ there is a unique decomposition given the functionals defined by each $g \in G$:

$$f = \sum_{h\in G} f(h)\delta_h \qquad \text{so that} \qquad f(g) = \sum_{h\in G} f(h)\delta_h(g) = f(g) \cdot 1 = f(g)$$

The key feature of this basis is that it acts like evaluation when taking the inner product. Define:

$$\langle f, g\rangle = \sum_{h\in G} f(h)\overline{g(h)}$$

Then $\{\delta_g\}_{g\in G}$ is an orthogonal basis and:

$$\langle f, \delta_{g_i}\rangle = f(g_i)\cdot\overline{\delta_{g_i}(g_i)} = f(g_i)\cdot 1 = f(g_i) = a_i$$

This is a special case of the Riesz representation theorem: every linear functional here is inner-product representable using these δ_{g_i}. This inner product can be thought as expressly chosen so that the collection δ_{g_i} forms an orthonormal basis, or conversely given this standard inner product, the indicator function will always be an orthonormal basis. By Pontryagin duality, $G \cong \hat{G}$, but this group isomorphism does not *a-priori* preserve the orthogonality of the basis given by G. The key insight of Fourier analysis is that it does!

Proposition 0.1.5: Character Form Orthogonal Basis: Fourier Basis

The characters $\widehat{G}$ form an orthogonal basis for $\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C})$. Normalizing by $1/\sqrt{|G|}$ forms an orthonormal basis.

***Proof*:**
If $i = j$:

$$\begin{aligned}\langle \chi_i, \chi_i\rangle &= \sum_{g\in G} \chi_i(g)\overline{\chi_i(g)}\\ &= \sum_{g\in G} |\chi_i(g)|^2\\ &= \sum_{g\in G} 1^2\\ &= |G|\end{aligned}$$

if $i \neq j$, Then as $\overline{\chi_j} = \chi_j^{-1}$ (as we are mapping to the unit circle), we get that $\chi' = \chi_i\chi_j^{-1}$ is a non-trivial character, so there is some h such that $\chi'(h) \neq 1$. Let

$$\langle \chi_i, \chi_j\rangle = S = \sum_{g\in G} \chi'(g)$$

Then:

$$\chi'(h)S = \chi'(h)\sum_{g\in G}\chi'(g) = \sum_{g\in G}\chi'(h)\chi'(g) = \sum_{g\in G}\chi'(h+g)$$

As $h + G = G$, this is just a re-arranging of terms, hence:

$$\chi'(h)S = \sum_{g\in G}\chi'(g) = S \qquad \text{so} \qquad S(\chi'(h) - 1) = 0$$

Now, by assumption $\chi'(h) \neq 1$, so $\chi'(h) - 1 \neq 0$, so as $\mathbb{C}$ is an integral domain it must be that $S = 0$, implying:

$$\langle \chi_i, \chi_j\rangle = 0$$

as we sought to show.

This inner product can be thought of as extracting the "phase" information.

Example 0.1.6: Phase-Space Basis

Let $G = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} = \{(0,0),(1,0),(0,1),(1,1)\}$. A character $\chi : G \to \mathbb{C}^*$ is determined by its values on the generators $g_1 = (1,0)$ and $g_2 = (0,1)$. Since both generators have order 2, we must have $\chi(g_1)^2 = \chi(g_1^2) = \chi(0,0) = 1$ and likewise for g_2. Thus, the character's values on the generators must be either 1 or -1. This gives $2 \times 2 = 4$ possible characters, which we can index by pairs $(k_1, k_2) \in \{0,1\}^2$.

The four characters χ_{k_1,k_2} are defined by their values on the generators:

$$\chi_{k_1,k_2}(1,0) = (-1)^{k_1} \qquad \text{and} \qquad \chi_{k_1,k_2}(0,1) = (-1)^{k_2}$$

From this, the value for any element $(a,b) \in G$ is $\chi_{k_1,k_2}(a,b) = (-1)^{k_1 a + k_2 b}$.

This gives the following character table, which explicitly lists the functions that form our new "basis":

$\hat{G}$	**(0,0)**	**(1,0)**	**(0,1)**	**(1,1)**
χ_{00}	1	1	1	1
χ_{10}	1	-1	1	-1
χ_{01}	1	1	-1	-1
χ_{11}	1	-1	-1	1

We can verify the orthogonality relation from proposition 0.1.5 directly. Let us compute $\langle \chi_{10}, \chi_{11} \rangle$:

$$\begin{aligned}
\langle \chi_{10}, \chi_{11} \rangle &= \sum_{g \in G} \chi_{10}(g)\overline{\chi_{11}(g)} \\
&= \chi_{10}(0,0)\chi_{11}(0,0) + \chi_{10}(1,0)\chi_{11}(1,0) + \chi_{10}(0,1)\chi_{11}(0,1) + \chi_{10}(1,1)\chi_{11}(1,1) \\
&= (1)(1) + (-1)(-1) + (1)(-1) + (-1)(1) \\
&= 1 + 1 - 1 - 1 = 0
\end{aligned}$$

Using the above example, let us change perspectives on how to interpret this change of basis. The function $f : G \to \mathbb{C}$ takes the "points" $g \in G$ and maps them (continuously) to the complex numbers, and those points are evaluated by the Riesz representation of f against the indicator functions that form an orthonormal basis. But we have just found another orthonormal basis for $\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C})$ against which we can take the inner product. We may thus think of the characters $\chi \in \hat{G}$ as points in their own right: points carrying "phase" information rather than the "location" information that the indicator functions carry. This motivates the following definition:

Definition 0.1.7: Fourier Transform (Gelfand Transform)

Let G be a finite abelian group with the discrete topology. Then the *Fourier transform* is:

$$\widehat{(-)} : \mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C}) \mapsto \mathrm{Hom}_{\mathbf{Top}}(\hat{G}, \mathbb{C}) \qquad f \mapsto (\chi \mapsto \langle f, \chi \rangle)$$

where the image of f is often written as:

$$\hat{f}(\chi) = \langle f, \chi \rangle = \sum_{g \in G} f(g)\overline{\chi(g)}$$

The name *Gelfand transform* comes from the fact that the Fourier transform is traditionally about real periodic functions, or Legesgue measurable functions. It wasn't until later where mathematicians like Folland and Gelfand generalized it to the context of locally compact abelian groups, where the above is the special case where G is finite.

Now, certainly as vector spaces $\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C}) \cong_{\mathbb{C}} \mathrm{Hom}_{\mathbf{Top}}(\hat{G}, \mathbb{C})$ mapping:

$$(g \mapsto \langle f, \delta_g \rangle) \mapsto (\chi \mapsto \langle f, \chi \rangle)$$

But does this map preserve the inner product, and more so can the convolution product of $\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C})$ be preserved given the appropriate choice of product on $\mathrm{Hom}_{\mathbf{Top}}(\hat{G}, \mathbb{C})$? The answer is yes, and can be thought of as the corner-stone result of harmonic analysis on discrete abelian groups:

Theorem 0.1.8: Fourier Reciprocity

The map given in definition 0.1.7 is a unitary $\mathbb{C}$-algebra isomorphism,

$$(\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C}), +, *) \cong (\mathrm{Hom}_{\mathbf{Top}}(\hat{G}, \mathbb{C}), +, \cdot)$$

with $\cdot$ in $(\mathrm{Hom}_{\mathbf{Top}}(\hat{G}, \mathbb{C}), +, \cdot)$ being point-wise multiplication.

Recall that unitary means $\langle \hat{f}, \hat{g} \rangle = \langle f, g \rangle$. In a conventional real analysis course:

- the $\mathbb{C}$-algebra homomorphism is called the *Convolution Theorem*,
- the unitary property is called *Plancherel's Theorem*,
- invertibility (i.e. bijectivity) is called the *Fourier Inversion Theorem*.

***Proof*:**

1. *1. Linearity:* Let $f, h \in \mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C})$ and $a, b \in \mathbb{C}$.

$$\begin{aligned}
\widehat{af+bh}(\chi) &= \langle af + bh, \chi\rangle && \text{by definition} \\
&= \sum_{g\in G} (af+bh)(g)\overline{\chi(g)} \\
&= a\sum_{g\in G} f(g)\overline{\chi(g)} + b\sum_{g\in G} h(g)\overline{\chi(g)} && \text{by linearity of sums} \\
&= a\langle f, \chi\rangle + b\langle h, \chi\rangle \\
&= a\hat{f}(\chi) + b\hat{h}(\chi)
\end{aligned}$$

Since this holds for all $\chi \in \hat{G}$, the transform is a linear map.

2. $\mathbb{C}$-algebra Homomorphism (The Convolution Theorem): We must show that the Fourier transform converts convolution into pointwise multiplication.

$$\begin{aligned}
\widehat{f*h}(\chi) &= \langle f*h, \chi\rangle \\
&= \sum_{k\in G} (f*h)(k)\overline{\chi(k)} && \text{by definition of inner product} \\
&= \sum_{k\in G}\left(\sum_{g\in G} f(g)h(g^{-1}k)\right)\overline{\chi(k)} && \text{by definition of convolution} \\
&= \sum_{g\in G} f(g)\sum_{k\in G} h(g^{-1}k)\overline{\chi(k)} && \text{swapping order of summation}
\end{aligned}$$

Let $\ell = g^{-1}k$, so $k = g\ell$. As k runs over G, so does ℓ. The inner sum becomes:

$$\sum_{\ell\in G} h(\ell)\overline{\chi(g\ell)} = \sum_{\ell\in G} h(\ell)\overline{\chi(g)\chi(\ell)} = \overline{\chi(g)}\sum_{\ell\in G} h(\ell)\overline{\chi(\ell)} = \overline{\chi(g)}\hat{h}(\chi)$$

Substituting this back into the main expression:

$$\begin{aligned}
\widehat{f*h}(\chi) &= \sum_{g\in G} f(g)\left(\overline{\chi(g)}\hat{h}(\chi)\right) \\
&= \hat{h}(\chi)\sum_{g\in G} f(g)\overline{\chi(g)} \\
&= \hat{h}(\chi)\hat{f}(\chi)
\end{aligned}$$

Thus, $\widehat{f*h} = \hat{f}\cdot\hat{h}$, proving the map is an algebra homomorphism.

3. Isomorphism (The Fourier Inversion Theorem): We show the map is invertible by constructing its inverse. Let $\mathcal{F}(f) = \hat{f}$. We claim the inverse is given by $\mathcal{F}^{-1}(\hat{f})(g) =$

$\frac{1}{|G|}\sum_{\chi\in\hat{G}}\hat{f}(\chi)\chi(g)$.

$$\begin{aligned}
\frac{1}{|G|}\sum_{\chi\in\hat{G}}\hat{f}(\chi)\chi(g) &= \frac{1}{|G|}\sum_{\chi\in\hat{G}}\left(\sum_{h\in G} f(h)\overline{\chi(h)}\right)\chi(g) && \text{by definition of } \hat{f} \\
&= \frac{1}{|G|}\sum_{h\in G} f(h)\sum_{\chi\in\hat{G}}\chi(g)\overline{\chi(h)} && \text{swapping order of summation} \\
&= \frac{1}{|G|}\sum_{h\in G} f(h)\sum_{\chi\in\hat{G}}\chi(gh^{-1})
\end{aligned}$$

The inner sum is an example of the *Second Orthogonality Relation* and is a standard exercise the reader should do. In particular, the sum of all characters evaluated at a group element (giving all powers of a certain root of unity) is $|G|$ if $x = e$ and 0 otherwise. Thus $\sum_{\chi\in\hat{G}}\chi(gh^{-1}) = |G|\delta_{g,h}$.

$$\frac{1}{|G|}\sum_{h\in G} f(h)(|G|\delta_{g,h}) = \sum_{h\in G} f(h)\delta_{g,h} = f(g)$$

Since we can recover f uniquely from $\hat{f}$, the map is an isomorphism.

4. Unitary Property (Plancherel's Theorem): Note that here is where it is important for the inner product to be normalized. Let us calculate the inner product for the given definition.

$$\begin{aligned}
\langle\hat{f},\hat{h}\rangle_{\hat{G}} &= \sum_{\chi\in\hat{G}}\hat{f}(\chi)\overline{\hat{h}(\chi)} \\
&= \sum_{\chi\in\hat{G}}\left(\sum_{g\in G} f(g)\overline{\chi(g)}\right)\overline{\left(\sum_{k\in G} h(k)\overline{\chi(k)}\right)} \\
&= \sum_{\chi\in\hat{G}}\left(\sum_{g\in G} f(g)\overline{\chi(g)}\right)\left(\sum_{k\in G}\overline{h(k)}\chi(k)\right) \\
&= \sum_{g\in G}\sum_{k\in G} f(g)\overline{h(k)}\sum_{\chi\in\hat{G}}\chi(k)\overline{\chi(g)} && \text{re-arranging terms} \\
&= \sum_{g\in G}\sum_{k\in G} f(g)\overline{h(k)}(|G|\delta_{g,k}) && \text{by Second Orthogonality} \\
&= |G|\sum_{g\in G} f(g)\overline{h(g)} \\
&= |G|\langle f,h\rangle_G
\end{aligned}$$

This proves *Plancherel's Theorem.* To make the map strictly unitary, one must normalize the transform. If we define the unitary transform $\mathcal{U}(f) = \frac{1}{\sqrt{|G|}}\hat{f}$, then it follows immediately that $\langle\mathcal{U}(f),\mathcal{U}(h)\rangle = \langle f,h\rangle$, completing the proof.

Example 0.1.9: Fourier Transform

Take $G = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ form example 0.1.6. Let us now compute the Fourier transform of a concrete function. Let $f = \delta_{(1,0)}$, so that $f(1,0) = 1$ and is zero elsewhere. This is a basis vector in the standard "position" basis. We find its coordinates in the character basis by computing $\hat{f}(\chi) = \langle f, \chi \rangle$ for each character.

$$\begin{aligned} \hat{f}(\chi) = \langle \delta_{(1,0)}, \chi \rangle &= \sum_{g \in G} \delta_{(1,0)}(g)\overline{\chi(g)} \\ &= 1 \cdot \overline{\chi(1,0)} + 0 + 0 + 0 = \chi(1,0) \end{aligned}$$

Using the character table, we can compute this value for each χ:

- $\hat{f}(\chi_{00}) = \chi_{00}(1,0) = 1$
- $\hat{f}(\chi_{10}) = \chi_{10}(1,0) = -1$
- $\hat{f}(\chi_{01}) = \chi_{01}(1,0) = 1$
- $\hat{f}(\chi_{11}) = \chi_{11}(1,0) = -1$

Thus, the original function $f = \delta_{(1,0)}$, which is a spike at one point in the "position" basis, has been transformed into a new function $\hat{f}$ on $\hat{G}$ with values $(1, -1, 1, -1)$.

Recall the inversion formula from the proof of theorem 0.1.8: $f(g) = \frac{1}{|G|} \sum_{\chi \in \hat{G}} \hat{f}(\chi)\chi(g)$ (here $|G| = 4$). Let's check the value at $g = (1,0)$:

$$\begin{aligned} f(1,0) &= \frac{1}{4}\left(\hat{f}(\chi_{00})\chi_{00}(1,0) + \hat{f}(\chi_{10})\chi_{10}(1,0) + \hat{f}(\chi_{01})\chi_{01}(1,0) + \hat{f}(\chi_{11})\chi_{11}(1,0)\right) \\ &= \frac{1}{4}\left((1)(1) + (-1)(-1) + (1)(1) + (-1)(-1)\right) \\ &= \frac{1}{4}(1 + 1 + 1 + 1) = 1 \end{aligned}$$

Now let's check the value at $g = (0,1)$, where we expect 0:

$$\begin{aligned} f(0,1) &= \frac{1}{4}\left(\hat{f}(\chi_{00})\chi_{00}(0,1) + \hat{f}(\chi_{10})\chi_{10}(0,1) + \hat{f}(\chi_{01})\chi_{01}(0,1) + \hat{f}(\chi_{11})\chi_{11}(0,1)\right) \\ &= \frac{1}{4}\left((1)(1) + (-1)(1) + (1)(-1) + (-1)(-1)\right) \\ &= \frac{1}{4}(1 - 1 - 1 + 1) = 0 \end{aligned}$$

Let us now connect this to Representation Theory. So far, we have established that for a finite abelian group G, the Fourier transform is a unitary $\mathbb{C}$-algebra isomorphism between the algebra of functions on G with convolution, $(\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C}), *)$, and the algebra of functions on the dual group $\hat{G}$ with pointwise multiplication. The representation theory perspective shows that Fourier analysis is the simplest case of a Artin-Wedderburn decomposition. The key is to view the $\mathbb{C}$-group algebra $\mathbb{C}[G]$ with the natural G-action given by left multiplication (convolution), namely $\rho : G \to \mathrm{GL}(\mathbb{C}[G])$

given by:

$$\rho(g) : f \mapsto g * f$$

From this viewpoint, the study of the $\mathbb{C}$-algebra $(\mathbb{C}[G], +, *)$ and the $\mathbb{C}[G]$-module $\mathbb{C}[G]$ is the same thing that is the study of the regular representation of G. Now, this representation can be broken down into the direct sum of 1-dimensional irreducible representations. Namely, for a finite abelian group, the set of irreducible representations is exactly the dual group: $\mathrm{Irr}(G) = \widehat{G}$ and so we get:

$$\mathbb{C}[G] \cong \bigoplus_{\chi \in \widehat{G}} V_\chi$$

where each V_χ is a 1-dimensional subspace corresponding to the character χ.

This decomposition is made concrete by the Fourier transform. The Fourier Inversion Formula, $f = \frac{1}{|G|} \sum_{\chi \in \hat{G}} \hat{f}(\chi)\chi$, is the literal expression of this decomposition. It writes the function f as a sum of components, where each term $\hat{f}(\chi)\chi$ is the projection of f onto the 1-dimensional irreducible subspace V_χ. Then the fourier coefficient $\hat{f}(\chi)$ is the coordinate of f in the irreducible component corresponding to χ.

Note that we can define projection functions $p_\chi = \frac{1}{|G|} \sum_{g \in G} \overline{\chi(g)} g$ that we can construct in $\mathbb{C}[G]$ that project onto the irreducible subspace V_χ. Convolving any function f with p_χ annihilates all other "frequency components", leaving the part of f that lies in V_χ.

Hence, the Fourier basis *the* unique basis that diagonalizes the natural action of the group on its own algebra of functions.

If G is a *non-abelian* finite group, the concept of a dual group $\hat{G}$ and a simple transform onto it ceases to be useful (it is now a set with no natural group structure). However, the representation theory perspective remains: the generalized Fourier transform (given by the *Peter-Weyl Theorem*) decomposes a function f on the group into a sum of *matrix coefficients* corresponding to these higher-dimensional matrix representations.

0.1.1 Generalization to Locally Compact Abelian Groups

What happens when we wish to analyze functions on abelian groups like $(\mathbb{R}, +)$, $(\mathbb{Z}, +)$, or the circle group $(S^1, \cdot)$? These are examples of *Locally Compact Abelian (LCA)* groups. Moving from the finite to the infinite setting requires us to replace our finite sums with integrals, but to integrate on an arbitrary group, we first need a notion of "volume" or "measure". The shape of the answer is already visible in the finite case: read the delta functions δ_h as *measures*, and every f becomes a sum of weighted point masses. What we want, then, is a measure on the group that the group operation cannot see. The essential new ingredient is the *Haar measure*. For any LCA group G, there exists a measure μ that gives a consistent way to integrate functions on the group.

Definition 0.1.10: Haar Measure

A *Haar measure* on an LCA group G is a measure μ on the Borel σ-algebra of G that is left-translation-invariant. That is, for any measurable set $E \subseteq G$ and any element $g \in G$:

$$\mu(gE) = \mu(E)$$

This measure is unique up to a positive scaling constant. For abelian groups, it is also right-translation-invariant.

See [2] for details on how to construct such a measure on any LCA that is unique up to scaling. This measure is the direct generalization of taking the order of the group.

Example 0.1.11: Haar Measure

1. For a finite group with the discrete topology, the Haar measure is the counting measure, and $\mu(G) = |G|$.
2. For $(\mathbb{R}, +)$, the Haar measure is the standard Lebesgue measure dx.
3. For $(S^1, \cdot)$, the Haar measure can be taken as $d\theta$.
4. For $(\mathbb{Z}, +)$, the Haar measure is again the counting measure.

Note that in the infinite case we can no longer represent points by indicator functions; the *Dirac delta distributions* take on that role instead. See [4] for the theory of distributions.

With the Haar measure, we can redefine our main objects of study by replacing every sum over G with an integral with respect to $d\mu$.

1. *The Group Algebra $L^1(G)$:* The space of all functions becomes too large. The natural replacement for the convolution algebra $\mathbb{C}[G]$ is the space of absolutely integrable functions, $L^1(G, d\mu)$. For two functions $f, h \in L^1(G)$, their convolution is defined by the integral:

$$(f * h)(g) = \int_G f(x)h(x^{-1}g)d\mu(x)$$

This space $(L^1(G), +, *)$ forms a Banach algebra.

2. *The Dual Group $\widehat{G}$:* It was proven that the correct codomain should be S^1. The dual group is the set of all *continuous* group homomorphisms into S^1.

$$\widehat{G} = \mathrm{Hom}_{\mathbf{Top}}(G, S^1)$$

The dual group $\widehat{G}$ is itself an LCA group. It is crucial to note that, unlike the finite case, it is rarely true that $G \cong \widehat{G}$. For example:

$$\widehat{\mathbb{R}} \cong \mathbb{R}, \qquad \widehat{\mathbb{Z}} \cong S^1, \qquad \widehat{S^1} \cong \mathbb{Z}$$

This highlights why it is essential to treat $\hat{G}$ as a distinct space. The theory that governs this relationship is *Pontryagin Duality*, which states that the double-dual is canonically isomorphic to the original group: $\widehat{\widehat{G}} \cong G$.

3. *The Fourier Transform:* The definition of the Fourier transform is the direct analogue of the finite case, replacing the sum with an integral. For a function $f \in L^1(G) \cap L^2(G)$, its Fourier transform is a function $\hat{f}$ on the dual group $\hat{G}$:

$$\hat{f}(\chi) = \int_G f(g)\overline{\chi(g)}d\mu(g)$$

Similarly, the inner product that defines the Hilbert space structure on $L^2(G)$ is:

$$\langle f, h \rangle = \int_G f(g)\overline{h(g)}d\mu(g)$$

With these definitions, the entire suite of theorems proven in the finite case carries over:

Theorem 0.1.12: Fourier Reciprocity on LCA Groups

Let G be an LCA group with Haar measure μ. Then there exists a unique Haar measure ν on the dual group $\widehat{G}$ such that the Fourier transform, appropriately normalized, is a unitary $\mathbb{C}$-algebra isomorphism. The main results are:

1. *The Fourier Inversion Theorem:* For a sufficiently well-behaved function f (e.g., $f \in L^1 \cap L^2$ such that $\hat{f} \in L^1$), the original function can be recovered by an integral over the dual group:
$$f(g) = \int_{\widehat{G}} \hat{f}(\chi)\chi(g)d\nu(\chi)$$
2. *Plancherel's Theorem:* The Fourier transform extends uniquely from $L^1 \cap L^2$ to a unitary isomorphism of Hilbert spaces, $\mathcal{F} : L^2(G, \mu) \to L^2(\widehat{G}, \nu)$. This means it preserves the inner product:
$$\langle \hat{f}, \hat{h} \rangle_{L^2(\widehat{G})} = \langle f, h \rangle_{L^2(G)}$$
3. *The Convolution Theorem:* For $f, h \in L^1(G)$, the Fourier transform maps convolution to pointwise multiplication:
$$\widehat{f * h}(\chi) = \hat{f}(\chi) \cdot \hat{h}(\chi)$$

***Proof*:**
This is the Pontryagin duality theorem; a full proof via the Gelfand transform belongs to the harmonic analysis of locally compact abelian groups and is not reproduced in this starred section.

The correspondence between the finite and general LCA settings can be summarized as follows:

Finite Abelian Case	**LCA Case**
Group G	LCA Group G
Order $\lvert G \rvert$	Haar Measure μ
Group Algebra $\mathbb{C}[G]$	Banach Algebra $L^1(G, \mu)$
Hilbert Space $\mathrm{Hom}(G, \mathbb{C})$	Hilbert Space $L^2(G, \mu)$
Summation $\sum_{g \in G}$	Integration $\int_G d\mu(g)$
Dual Group $\widehat{G} = \mathrm{Hom}(G, \mathbb{C}^\times)$	Dual Group $\widehat{G} = \mathrm{Hom}_{\mathbf{Top}}(G, S^1)$
Inner Product $\sum f(g)\overline{h(g)}$	Inner Product $\int f(g)\overline{h(g)}d\mu(g)$
Fourier Transform $\sum f(g)\overline{\chi(g)}$	Fourier Transform $\int f(g)\overline{\chi(g)}d\mu(g)$

Further interesting facts can be found here: https://mathoverflow.net/questions/37021/is-fourier-analysis-a-special-case-of-representation-theory-or-an-analogue

0.1.2 The Algebraic-Geometric Meaning of the Dual Group

In algebraic geometry, a fundamental principle is that the geometry of a space can be recovered from its algebra of functions. For an affine space k^n, the corresponding algebra is the polynomial ring $k[x_1, \ldots, x_n]$. The points of the space correspond to the maximal ideals of this algebra. Specifically, for any point $p = (a_1, \ldots, a_n) \in k^n$, the evaluation map:

$$\mathrm{ev}_p : k[x_1, \ldots, x_n] \to k \qquad f \mapsto f(p)$$

is an algebra homomorphism whose kernel, $m_p = (x_1 - a_1, \ldots, x_n - a_n)$, is a maximal ideal. The quotient $k[x_1, \ldots, x_n]/m_p \cong k$ recovers the value of the function at that point.

We have established a similar setup for a finite abelian group G, with the convolution algebra $(\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C}), *, +)$ taking the role of the function algebra. A natural question arises: can we recover the "points" of our space from the maximal ideals of this algebra? Let's first attempt a direct analogy. We might assume the points are the elements of the group G itself. For a point $g_i \in G$, we can define the point-evaluation map, which is a linear map:

$$\mathrm{ev}_{g_i} : \mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C}) \to \mathbb{C} \qquad f \mapsto f(g_i)$$

The kernel of this map is the vector subspace of functions that are zero at g_i. However, for this subspace to be an ideal, it must be closed under convolution. Let $f \in \ker(\mathrm{ev}_{g_i})$ (so $f(g_i) = 0$) and let h be any function in our algebra. Consider their convolution:

$$(h * f)(g_i) = \sum_{k \in G} h(k) f(k^{-1} g_i)$$

Just because $f(g_i) = 0$ does not imply that the sum on the right-hand side is zero. The value of f at other points, $f(k^{-1} g_i)$, can be non-zero. Therefore, the kernel of the point-evaluation map is *not* an ideal with respect to convolution. This implies the points of G do not correspond to the maximal ideals of its convolution algebra.

The key insight is that the correct "geometric points" for the convolution algebra are not the elements of G, but the characters of the dual group $\widehat{G}$.

Proposition 0.1.13: Characters and Maximal Ideals

Let G be a finite abelian group. The maximal ideals of the convolution algebra $(\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C}), +, *)$ are in bijective correspondence with the characters $\chi \in \widehat{G}$.

***Proof*:**

For each character $\chi \in \widehat{G}$, define a Fourier evaluation map:

$$\widehat{\mathrm{ev}}_\chi : \mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C}) \to \mathbb{C} \qquad f \mapsto \hat{f}(\chi)$$

This map takes a function and returns its Fourier coefficient at the frequency χ. By 0.1.8 $\widehat{\mathrm{ev}}_\chi$ is a $\mathbb{C}$-algebra homomorphism. Since $\widehat{\mathrm{ev}}_\chi$ is a surjective algebra homomorphism onto a field ($\mathbb{C}$), its kernel, which we denote by I_χ, is a maximal ideal:

$$I_\chi = \ker(\widehat{\mathrm{ev}}_\chi) = \{f \in \mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C}) \big| \hat{f}(\chi) = 0\}$$

By the First Isomorphism Theorem for rings, we have:

$$\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C})/I_\chi \cong \mathbb{C}$$

This shows that every character $\chi \in \widehat{G}$ defines a maximal ideal. Conversely, After applying the Fourier transform we get the usual multiplication and ring structure, and then an approach similar to the Gelfand-Kolmogorov theorem gives the desired result.

For the infinite case, the maximal ideals of the Banach algebra $L^1(G, \mu)$ arise in this same way from homomorphisms into $\mathbb{C}$ (see the *Gelfand-Mazur Theorem*).

Thus, with the convolution algebra $\mathbb{C}[G]$ as our primary object, the natural geometric space that emerges is not the group G itself, but its Pontryagin dual, $\widehat{G}$. The Fourier transform is the mechanism that allows us to evaluate the functions of our algebra at the points of this "true" underlying space. This perspective, generalized by Gelfand, is the foundation of non-commutative geometry.

Concepts	**Classical Algebraic Geometry**
Algebra	Polynomial Ring $(k[x_1, \ldots, x_n], \cdot)$
Product	Polynomial Multiplication
Geometric Space	Affine Space k^n
A "Point" p	A tuple $(a_1, \ldots, a_n) \in k^n$
Evaluation Map	$\mathrm{ev}_p : f \mapsto f(p)$
Maximal Ideal for p	$\ker(\mathrm{ev}_p) = (x_i - a_i)$

Concepts	**Harmonic Analysis on G**
Algebra	Group Algebra $(\mathbb{C}[G], *)$
Product	Convolution
Geometric Space	The Dual Group $\widehat{G}$
A "Point" p	A character $\chi \in \widehat{G}$
Evaluation Map	Fourier "Evaluation" $\widehat{\mathrm{ev}}_\chi : f \mapsto \hat{f}(\chi)$
Maximal Ideal for p	$\ker(\widehat{\mathrm{ev}}_\chi) = I_\chi$

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